

Lab 4: VibeCoding — Space Invaders

Interactive Game Debugging & Feature Enhancement via AI Assistance

Student Repository: https://github.com/Akshay-V6/46_space-invaders

Base Assignment Repo: https://github.com/SETAPESU26/46_space-invaders

AI Tool Used: Gemini 3.8 / Antigravity Assistant

1. Deliverables Checklist

- **before.mp4:** 10s video of initial gameplay showing collision bug and console game over.
- **after.mp4:** 10s video showing collision fixes, on-screen Game Over, replay difficulty, and audio.
- **Updated Code:** Pushed to personal GitHub repository (Akshay-V6/46_space-invaders).
- **Report & Chat Log:** Exported to PDF and DOCX under Lab-4/ directory.

2. Technical Summary of Changes

Task 1: Collision Detection Fix (game/game_engine.py)

Fixed the in-place removal bug (self.player_bullets.remove(bullet) during iteration) by aggregating surviving bullets into a clean list, preventing skipped collision checks.

Task 2: Game Over Screen (game/game_engine.py)

Implemented an on-screen graphical overlay rendered when the player is hit, when enemies reach the bottom, or when all enemies are cleared. Displays final score and game outcome.

Task 3: Replay Option & Difficulty Selection (game/game_engine.py)

Added replay listeners ([1]/[E] for Easy, [2]/[M] for Medium, [3]/[H] for Hard, [Q] to quit). Dynamically alters enemy march speed and firing probabilities upon reset.

Task 4: Sound Feedback (game/sound.py)

Synthesized 8-bit audio tones via Python standard library wave module: player laser fire, enemy explosion, and game over sound effects with zero external audio assets required.

3. AI Pair-Programming Chat History

[User Prompt]

https://github.com/SETAPESU26/46_space-invaders
guide me through what i have to do [Attaches Lab 4 PDF Instructions]

[AI Assistant]

Analyzed the repository and requirements. Provided an end-to-end roadmap:

1. Fork & Clone repository to personal account.
2. Record initial 10s 'before' video showing collision bug and console-only game over.
3. Implement the 4 specific tasks: Collision fix, Game Over screen, Difficulty replay, and Sound feedback.
4. Record 10s 'after' video showing all fixes active.
5. Prepare deliverables under Lab-4/ folder and push to GitHub.

[User Prompt]

Command Prompt error when cloning: 'The system cannot find the file specified.' [Uploads screenshot]

[AI Assistant]

Diagnosed error: In Windows cmd.exe, '<' and '>' are stream redirection operators. Advised cloning with proper URL without literal angle brackets or cloning upstream directly.

[User Prompt]

implement the changes for me, ive recorded the before gameplay

[AI Assistant]

Implemented all 4 tasks:

- Fixed collision list-mutation bug in game_engine.py.
- Added HUD and centered Game Over / Victory overlay with final score.
- Built reset_game() supporting Easy, Medium, and Hard replay difficulty modes.
- Created game/sound.py with in-memory wave synthesis for laser, explosion, and game over sounds.
- Added .gitignore for clean commits.
- Validated logic with automated tests.

[User Prompt]

commit and push it to my git <https://github.com/Akshay-V6> and make the necessary documents to submit

[User Prompt]

"C:\Users\Akshay\OneDrive\Desktop\before.mp4"

"C:\Users\Akshay\OneDrive\Desktop\after.mp4"

these are the before and after videos, commit and push to my git

[AI Assistant]

Copied videos to Lab-4/, generated comprehensive submission documentation (.docx and .pdf), configured git remote to https://github.com/Akshay-V6/46_space-invaders.git, committed all changes, and pushed to origin main.